

Visualizing Client Interactions in B2B Context: Design Explorations

Tenzin Yeshi*
Alice Y. Lee*
Robert J. Moore
Guang-Jie Ren
tenzin.yeshi@ibm.com
alice.lee@ibm.com
rjmoore@us.ibm.com
gren@us.ibm.com
IBM Research
USA

ABSTRACT

In business-to-business (B2B) commerce, the provider organization's collective understanding of the client organization is distributed across the communications among a multitude of individuals. However, the substance of many of these communications typically goes unrecorded even when using standard Customer Relationship Management (CRM) tools and therefore their value remains largely untapped. In an attempt to capture these valuable client interaction data, we built an intuitive data capture tool that fits in with sellers' client-facing workflows. But despite giving positive feedback on our prototype, our study participants informed us that the company was transitioning to a new CRM system, an industry standard. Therefore, we pivoted to creating visualizations for client insights. Based on extensive feedback from our participants, we designed four novel B2B interaction visualizations: 1) Interactions, 2) Timeline, 3) Topics and 4) Network. In this paper, we report on our design exploration for using CRM data to visualize client relationships in B2B context and discuss the challenges we encountered.

CCS CONCEPTS

• **Human-centered computing** → **Scenario-based design**; • **Information systems** → *Data analytics*; • **Applied computing** → **Enterprise information systems**.

KEYWORDS

client interactions, interaction visualization, topic visualization, timeline visualization, network visualization

ACM Reference Format:

Tenzin Yeshi, Alice Y. Lee, Robert J. Moore, and Guang-Jie Ren. 2023. Visualizing Client Interactions in B2B Context: Design Explorations. In *Extended CHI EA '23*, April 23–28, 2023, Hamburg, Germany.

*Both authors contributed equally to this research.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3573860>

Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23), April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3573860>

1 INTRODUCTION

In business-to-business (B2B) contexts, clients are the most important players as their requirements are the sources of value creation. Client requirements in B2B contexts are complex and often more than just ordering of existing products and services. They can be communicated in documents such as annual reports and press releases, but a lot of them are expressed through direct interactions between individual clients and providers. As such, it is crucial for B2B companies to capture and analyze client interactions in order to understand client needs and make compelling value propositions.

Usually, client interactions are collected by client-facing teams and sharing is limited to local and adjacent teams. When team members leave, the information and insights often get lost. With the adoption of Customer Relationship Management (CRM) systems, it is hoped that information about client interactions is stored centrally, minimizing the impact of personnel changes. The reality, however, is far from ideal. In this paper, we report our observations and design explorations of client interactions data in B2B contexts.

2 BACKGROUND

The motivation for this project was to enhance the work of B2B sellers through new collaboration tools. We learned that much of their communication with clients is not documented in any shareable way. In evaluating seller's existing tools, we discovered that CRM systems had very little information about interactions at the early stages of a client engagement. Indeed, most information appeared to be captured only after specific offerings were identified to facilitate the order-taking process.

To enable B2B sellers to gain more value from their client interaction data, we designed a tool for capturing this data, sharing it across sellers and generating client insights. Our goal was to capture information on interactions with clients earlier in the engagement life cycle, when clients are explaining their problems and requirements but a solution has not yet been found. Such upstream interaction data, once captured, could later be imported to a standard CRM system for sharing. With this in mind, we built a

prototype that enabled users to capture interaction data through a conversational agent on a mobile device and to view and enhance the data through a web interface. More specifically, the user may travel to different locations to meet with clients and while on the go, waiting at an airport or in a taxi, they could interact with a conversational agent through the standard short message system (SMS). The user could both record client-related details (e.g., “I met with Jane Smith today and talked about advantages of hybrid cloud”) and inquire about them (e.g., “who else has talked to Jane Smith recently?”). And when users return to their desktop computers, they could enrich records with more detail, review past records, share them and gain insights through data visualizations.

To articulate user needs, we interviewed multiple B2B sellers and sales managers across multiple industries. Our study participants validated the importance of capturing client interaction data and confirmed that it was currently absent from CRM tools as they were not integrated into their daily work practices. Creating records of client interactions was unnecessary extra work for them because the records were not shared across users but used only to track their individual productivity. This is a classic reason that collaboration applications fail: due to “the disparity between who does the work and who gets the benefit” [1]. Indeed, if the sellers log those conversations in a system that is reviewed by their management, they will receive pressure to turn that conversation into a sales opportunity.

Our participants also noted that it is not always useful to capture client interaction data. They explained that not all early conversations about clients represent business opportunities. Some of those interactions may well be the realization that there is no identifiable solution. In addition, clients tell sellers some things in confidence, such as criticisms of the seller’s company or intentions to go with a competitor. Capturing such conversations in a shared repository would violate the client’s hard-earned trust in the seller, and if entered into a shared system, they would want control over who has access to these records.

Nonetheless our participants suggested that they could receive benefit from client interaction data if it generated insights useful for their work, enterprise sales. They explained that CRM systems tend to lack features for these purposes, especially at the individual levels on both client and provider sides. Indeed, CRM systems are initially built for Business-to-Consumer (B2C) companies which mostly deal with individual consumers and sellers tend to operate sales activities within the CRM tools. They capture limited information about client requirements and have little need to analyze interactions for insights. In comparison, B2B companies interact with a host of individuals and teams in the client organization and the membership on both sides changes on a regular basis.

However, toward the end of our user study, we also learned that in the following year, the sales organization would be mandated to use new CRM tool, an industry standard, and would be required to capture more client interaction data in it. This would make our capture tool redundant. At this point, we pivoted our project to abandon the capture and sharing functions in our design and to focus on generating insights through visualizations of the client interaction data. Although we ultimately abandoned our prototype capture tool, it nonetheless facilitated the interactions with our

study participants and our understanding of their work and real needs.

3 PROJECT DETAILS

Through several follow-up user studies on interaction visualization, we identified four user goals:

- **Interactions** - As an account team, we want to see who interacted with who and how much (by individual) between our company and a client account, including number of interactions, most active individuals and high-level topics discussed
- **Timeline** - As an account team, we want to see how the client relationship has evolved over time (by meeting) between our company and a client account, including the meeting size, topics, opportunities, lines of business
- **Topics** - As an account team, we want to explore what was talked about and how much (by topic) across a client account or industry, to understand the client’s interests and what products and services are being discussed
- **Network** - As an account team, we want to discover the non-obvious connections among a wider population of individuals (across accounts), among both client contacts and colleagues, in a social network graph based on past interactions or organizational roles

It is worth noting that while each of the user goals focuses on a distinctive perspective of client interactions, they also interrelate with each other. For instance, Topic visualization provides a holistic view of topics across interactions but some of the topics are also shown in Interaction and Timeline visualizations as a secondary dimension. The same applies to Interaction and Network visualizations; both offer organization-level insights about social connections even though the former is more structured and the latter more fluid.

3.1 Interactions

Business accounts consist of various individuals and teams from both organizations. Members’ involvement and interactions change over time as an account goes through different phases. This web of players interacting about different topics over time is often presented in simple data tables. Some CRM tools have dashboards with aggregated summaries and breakdowns. While data tables are efficient in retrieving logs and dashboards offer an overview, they provide few meaningful insights.

Our target users wanted information beyond logs. More specifically, they want to see who interacted with whom the most, how much or how often they interacted, and what topics they discussed. They need a tool that they can use to understand these relationships and patterns with ease, which current CRM systems do not offer. To meet this need, we started by exploring concepts to visualize interactions at the organizational level as well as between individuals.

We evaluated several visualizations that allow an understanding of relationships between the two sides and their patterns over time, including network diagrams, Sankey, and Alluvial. Network diagrams are ideal for illustrating network architecture, and Sankey and Alluvial diagrams are well suited for representing the distribution and flow of datasets. While these visualizations work well to plot correlations between variables, these off-the-shelf diagrams

fall short and lack the flexibility to meet our requirements. As a result, we decided to design a bespoke visualization that users can intuitively comprehend.

3.1.1 Solution. The proposed solution was to compose a layout that leverages left-to-right visual scanning to orient our users (see Figure 1). The left panel is for the provider, where the user’s colleagues are, and to the right is the client organization. Each node represents a member, and they are vertically stacked in an order that considers not only the number of interactions but also the organizational hierarchy. With activity and importance, we can crop up an accurate list of members. And we fold members that are less relevant, resulting in a concise list with options to expand.

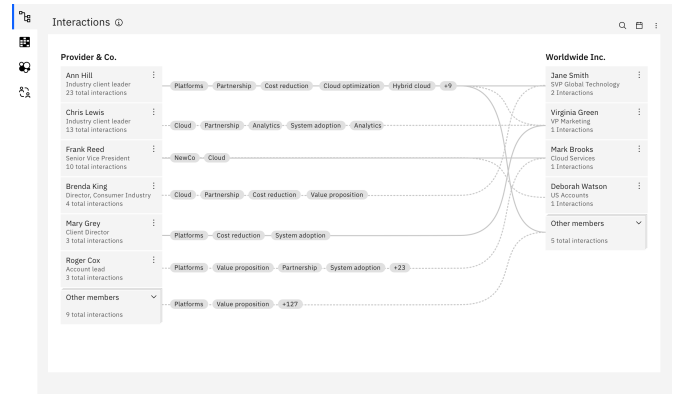
These nodes are connected with lines that represent interactions, creating a network that is of the account. This gives our users an idea of their account’s state at the organization level and an isolated interaction flow reveals the relationship between two select individuals. The lines are treated in different styles, stroke thickness and styles, to illustrate the nature and strength of interaction. This allows users to understand relationships better at a glance, solving one of the pain points users face with traditional data logs.

In addition, we added tags for high-level topics associated with interactions. The tags allow users to scan easily and inform what interacting members talked about. These tags not only label interactions but are filterable, meaning the user can select a tag to surface who else talked about the same topic. These topic tags can also be used to navigate to the Topics tab for further investigation.

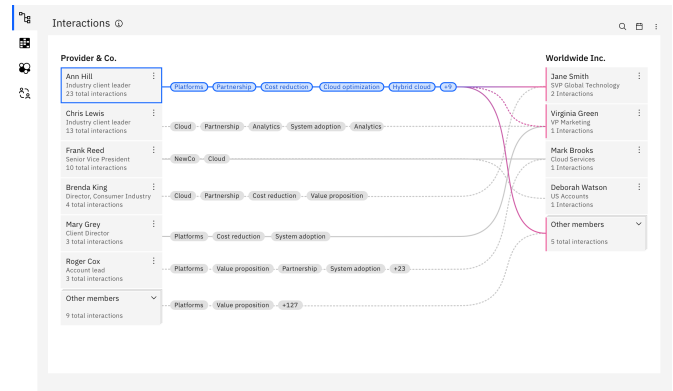
Similarly, an interaction flow (i.e., the tiles, the line connecting them, and the attached tags) has a brush feature. As seen in Parallel Coordinate and Sankey diagrams, it isolates the selected section aiding users to effectively focus on a specific flow they are interested in and fade out the rest. From an interface standpoint, we kept it uncluttered and easy to interact with by employing such interactive apparatuses mentioned above.

3.1.2 Challenges. We encountered some challenges as we designed interaction visualization. First, connections vary in importance and relevance. For instance, a connection can represent individuals who simply attended the same meeting or a critical one-on-one interaction between two executives. We had to find ways to make these types distinct and include high-level topics associated with interactions so that users can see what their colleagues talked about with client contacts (e.g., “Who spoke with client contact Jane Smith about – topic the most?”)

As with any visualization, design is as good as its data sets. But the current CRM systems are sparsely populated, and the input is inconsistent. Early on, we realized we must work with limited data set. Our participants provided us with a sample set of data without any confidential client information to use for the prototype. We decided to use only the most essential information such as attendees, meeting date and time, and high-level topics discussed. Working with that limitation helped us narrow our focus to only what is essential for our users. In other words, the visualization would be meaningful no matter the size of the organization, account, or content of the interaction log.



(a) Interactions visualization design screen without selection



(b) Interactions visualization design screen with selection

Figure 1: Interactions visualizations design screens

3.2 Timeline

While the Interaction visualization offers insights into the client relationships, it is not intended to provide those insights along the time dimension. Our users expressed the need to be able to see the history of their client relationships so as to project or identify areas of focus in the future; this meant we needed to find a way to show how the records were changing as the relationship progressed.

The Timeline visualization was created to offer another way for our users to look at the data and reveal how a client relationship changes over time. Instead of a simple linear timeline, our design approach was to enrich the visualization with additional layers of information. With this timeline visualization, our users would be able to see the health of a client relationship and high-level topics discussed over the course of an account.

There are various existing visualizations for records of a user’s actions or interactions. However, the majority of these visualizations, such as Microsoft’s timeline control in Power Apps, plot records across a single dimension. Although these timelines are simple and familiar to most users, time being the only dimension limits the potential of it providing more meaningful intelligence. There are a few interactive products, like Brightmetrics’ Agent Status Timeline Chart [5] and Salesforces’ TimeWarp [2], that allow users to get more information on a record through selecting, hovering, and

filtering. However, the single dimensional design would not allow us to illustrate the bigger picture of a client's journey and meet our users' needs.

3.2.1 Solution. Recognizing these limitations, we decided to eliminate the constraints by adding a vertical dimension and incorporating more sophisticated user interactions. We created a design that layers three different visualizations: a bubble chart and heatmap graph behind a timeline (see Figure 2).

A bubble chart shows the relationship between three data variables, each corresponding to either the x-axis, y-axis, or bubble size. In our design, the x-axis was time, the y-axis was interaction topics, and the bubble size represented interaction sizes or number of people involved. A heatmap graph shows the density of data, and in our design, this meant showing the density of the interaction data over distinct time segments. Users would also be able to manipulate the dimensions and change the layout as they see fit. The result was a multi-layered visualization that is easy to understand and offered insights our users needed.

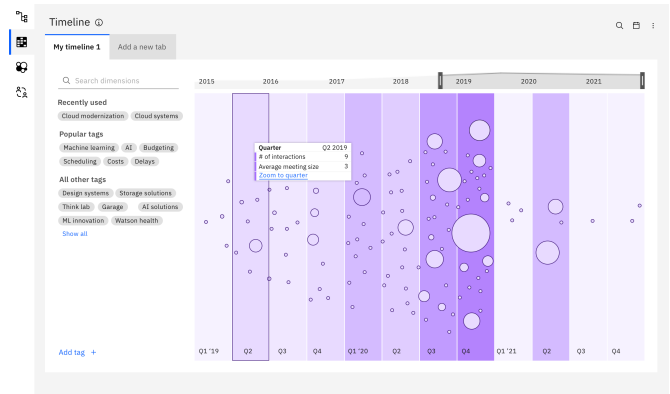
The y-axis of the bubble chart was the only categorical variable with each topic corresponding to its own row, and users would be able to select the topics they wanted displayed. But this came with its own unique challenge of determining which default view would be best for the graph prior to the user making any selections. Without a vertical axis, interactions that took place on the same day would have the same horizontal position and no system for vertical placement, making it difficult for users to read the visualization. For our final design we decided to place them in random vertical positions so to make sure the bubbles did not overlap when no y-axis was added yet.

With the heatmap background, users would see the trend in their interactions even without the y-axis set. A heatmap is set in sequential scales, leveraging the full range of the color to represent the spread of data. This subtle technique provides clear visual information at a glance. The combination of the heatmap with the simple horizontal timeline and bubble chart visualizes not only an easily interpreted representation of records over time but also the changes in frequency across discrete units of time.

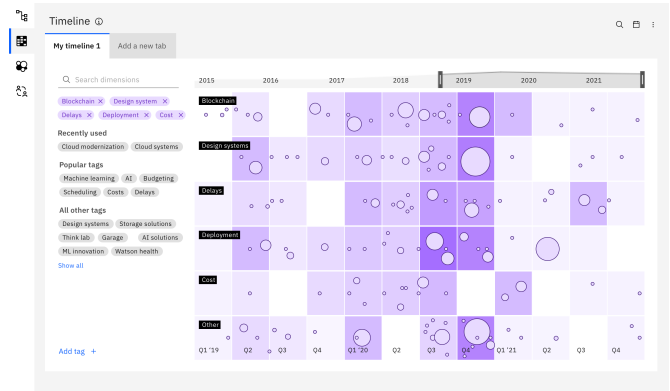
Our final design places the layered graphs onto a single base x-axis of time to tie them together. All records are placed along that axis and are represented by individual bubbles. The heatmap background is sectioned by either quarters, months, or weeks depending on the users' preference. Users can adjust this time window using the "zoom bar" at the top of the graph. Each section is a cell of the heatmap, with the background color determined by the number of interaction records in that section. Users can select different topics to add to the vertical axis of the graph, and the layout will update to add rows to the heatmap and rearrange the bubbles accordingly.

3.2.2 Challenges. With our decision to explore beyond the traditional timeline design, we needed to make sure the solution we came up with wasn't too complicated for our users. Oftentimes, larger data visualizations are crowded with an overload of information, so we needed to make our visualization purposeful and intuitive to interact with.

Adding vertical space to a timeline gave us more freedom but also gave us the challenge of coming up with a system for the new layout. As mentioned before, we needed to figure out how to



(a) Timeline visualization with no topics



(b) Timeline visualization with topics

Figure 2: Timeline visualization design screens

organize the interactions vertically when no y-axis was selected. Other timelines that extend their vertical space use a standard static vertical axis to plot elements based on that metric or value, but since our y-axis is mutable, bubbles would have to be placed randomly.

Another interesting problem we came across in later iterations was dealing with duplicates as an interaction could cover multiple topics. If multiple topics are tagged for an interaction, the bubble would need to show up in each of those topics' rows, but having the same bubble appear repeatedly would be an inaccurate visual representation of the number of records. To deal with this, we decided that bubbles would only appear once across the whole graph, and if there were multiple topics for that interaction, duplicate bubbles would appear as faded out with dashed outline in the other rows. These copies would also only be visible and highlighted upon hovering or selecting a bubble.

3.3 Topics

The topics visualization is another way for users to see the details of their interactions, focusing on what was talked about and how much across a client account or industry, including general topics and specific products. Our users need to know what is being discussed in any client interactions to figure out what are the best

solutions for the problems they have and to see if there is any specific area or functionality that the clients are interested in. In most client relationships, needs and solutions transform over time, which makes keeping track of the ideas discussed crucial to our users. For this visualization, our goal was to find and use the most relevant these technologies to provide users with the insights they need and present them visually with a clear and simple design that users can use to identify important business content and find opportunities to enhance client conversations.

There are various clustering methods that exist, but for our purposes of topic mining, we found K-means and Latent Dirichlet Allocation (LDA) to be the most useful. K-means clustering can be a soft or hard clustering, meaning it can provide multiple or just one topic per document. LDA is a soft-clustering algorithm that generates “topics,” or clusters of related words that occur in similar contexts across a corpus of documents. In our use case, such documents would be the descriptions of particular interactions with clients. The popular gensim package for LDA [3] provides a visualization of topics and the words in each topic after the analysis is finished. This end visualization uses a bubble chart to show the size of each topic group and a bar chart to show the frequency of each of the words in the topic.

3.3.1 Solution. Our solution originally combined the word cloud with the LDA model’s bubble visualization and added an extra layer of interaction and magnification for users to easily change the granularity of the topics, products, and ideas. After running the interaction records through the LDA algorithm, the results would be presented in a similar fashion as the gensim package. However, our design removes the two axes that the bubbles are placed on, as the topic distance is not relevant to our user goals and places the topics so that they are not overlapped in the space. It also changes the parameters, using bubble size to represent the number of interactions in the topic instead of the number of terms, which is what LDA uses to calculate bubble sizes.

The “data dimensions” panel allows users to filter which topics they see by selecting the terms they are interested in, and this can help them narrow down on whether the topics they select are being discussed and to what length.

Although we initially considered using the word cloud, a simple graphic that shows the frequency of words by changing their sizes, it was left out in the final iteration because we concluded that it is a more simplistic analysis, and the crowding of many words in varying sizes would over-complicate the screen.

3.3.2 Challenges. The biggest challenge for us was using NLP to provide the topic breakdown. Before starting the design process, we needed to investigate analysis methods in order to have the data for the design. We consulted with data scientists to decide which method to use, but there was no perfect match. Current NLP methods do a great job at clustering words and finding distance between them, but these methods do not always provide easy to understand results. The distance between topic clusters and the methodology behind how the clusters are created are not commonly understood and providing this information to our users would not help accomplish the end goal of understanding the topic distribution in their interactions. NLP methods also do not provide labels for the generated clusters, so these would need to be provided by a

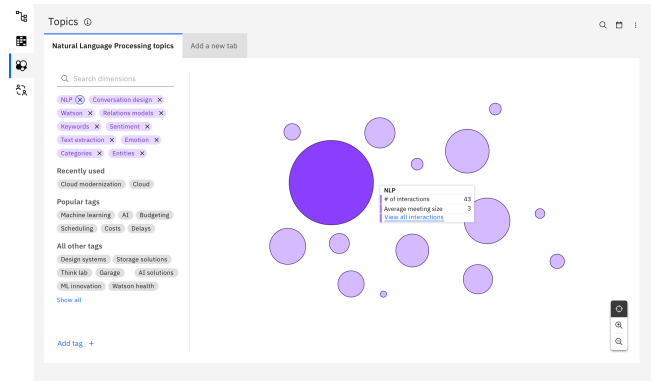


Figure 3: Topics visualization design screen

human down the line. We decided to leave off labels for our final design because of our lack of data, but we listed the terms included in each topic when selected so that users could see the relevant words.

3.4 Network

Our main objective for this visualization was to allow our users to discover the non-obvious connections among a wider population of individuals across accounts. Users would see who, among both client contacts and colleagues, is connected to whom in a variety of ways. This social network graph would be based on past interactions, organizational roles, career history, and their personal relationships.

In the social networks found on popular sites like LinkedIn and Facebook, the connections between people are based on a single dimension, the degree of association. Their associations are based solely on mutual contacts and fail to demonstrate the nature and strength of the relationships. In B2B context, establishing trust and maintaining relationships with clients is crucial in meeting business goals and future opportunities. Considering the number of individuals involved in such business dealings, finding and connecting with the point of contact with the most potential for a strong connection would be highly valuable for sellers.

Conceptually we approached this visualization with the user as the starting point, target contact as the destination, and connections between the two as routes. Hence, we looked at navigational tools for inspiration. Offering different route options based on users’ criteria as the output differentiated our visualization. We also looked at other social network visualizations, but these were too complex and crowded to use. Making sure our visualization didn’t have the same problems, we focused on clarity and salience of each component in our design.

3.4.1 Solution. The solution we designed was a social network visualization that is easy to interact with when finding the best connection paths to a target contact. Keeping the layout and component design consistent across the tabs helps our users to easily orient themselves. The left panel displays contextual details (see

Figure 4a). And with filter and search capabilities, we avoided making the visualization overcrowded and offered multiple ways to drill down.

To enrich the visualization, we resorted to “pre-attentive attributes” [4] like different line styles, stroke thickness, varying node sizes, and the use of colors. We assigned the primary color (blue) for the provider and a secondary color (magenta) for client contacts, and they are sized according to the individual’s relevance and activity. Links are color-coded similarly, and they graduate to indicate cross-organizational connections and dashed lines are for intra-organization connections. And their thickness represents connection strength.

The visualization plots all connections, colleagues and client contacts, for a target contact and contextual details on the left panel offer sellers the required background information. Their career history doubles as a filter functionality and can be used to segment their connections by different periods of their career. By adding time as an additional dimension to slice the network, sellers can sieve connections and discover paths efficiently.

Recommended connections paths are brushed and the list is displayed on right panel ranked in the order of relevance based on their past interactions, organizational roles, career history, and their personal relationships (see Figure 4a). This allows sellers to easily identify the right person to connect with and ultimately help with carrying out business deals successfully.

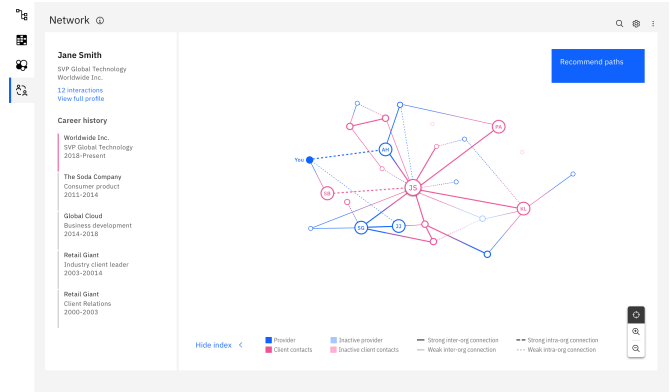
3.4.2 Challenges. Network diagrams primarily employ just two elements, nodes and links, and have limited data capacity. Most existing social network services rely on this simple design, but when it is scaled beyond a certain number of nodes, it becomes too dense to read. Despite its limitations, this visualization was still the most suitable for our purpose, so we decided to add filters and navigational options to reduce the pool of connections for the final diagram.

One design challenge was adding more visual information and enriching the otherwise rudimentary diagram. Since we are working with limited elements, we had to explore various additional variables that we can layer. Working with our study participants, we also came across cases of members moving from one account to another and even switching organizations. Visually representing such users added further complications.

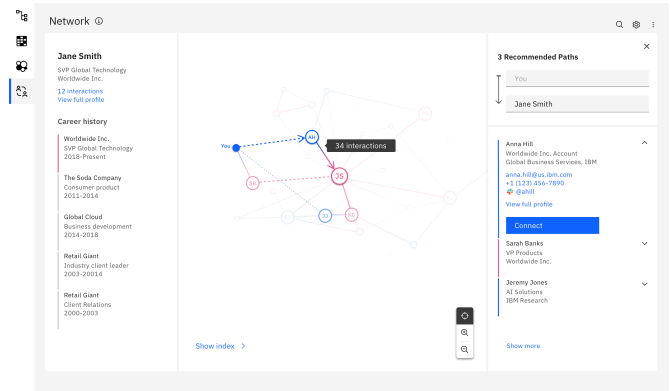
4 DISCUSSION

Through our experimentation and interviews with study participants, we concluded that a single type of visualization was inadequate for capturing the complexity of a relationship between B2B organizations and for meeting the needs of sellers in doing their jobs. We offer reflections on each visualization: Interactions, Timeline, Topics and Network.

First, with the Interactions view, the scope was to visualize two organizations, but we realized that may not be the only use case. We could foresee cases where there are more than two parties involved. Although cases like this are not common, we would need to consider scalability in our design decisions. In addition, a question our sellers are also interested in is “how did the meeting go?” It is a rather simple question, an obvious one, for a human to answer. But for the



(a) Networks visualization with full network



(b) Networks visualization with target person selected

Figure 4: Networks visualization design screens

visualization to address this we would need to collect sentiments associated with interactions. That knowledge is tribal and is not captured anywhere. Even if we do manage to collect sentiments associated with an interaction, we could see ourselves running into issues around reliability and subjectivity. Nevertheless, our users validated that such insights could be very valuable and it is an interesting area for future exploration.

Second, creating the Timeline visualization was a unique process as the goal was not just to present the data, but to generate insights from the analytics. We had to determine which aspects of the data and which analytics information were most important to incorporate in the design. This Timeline design is an informative and relatively simple visual that could be used to see patterns in datasets and see the trends based on various combinations of filters on the vertical axis. Specifically, it shows the evolution of interactions over time and provides insight on the direction of business.

The vertical arrangement of the bubbles may lead to some difficulty with use as there is no system for determining their positions. One possible solution would be to extract interaction sentiments and place them in negative, neutral, or positive sections. But this would require complicated sentiment analysis, which has not been possible at this stage, nor has user intervention by recording sentiments for each record.

Third, the design of the Topics visualization is a simple display of bubbles that is easy to understand as it is a basic representation of groupings and frequency, and it can be a useful tool to see the trends in discussions. The mutable granularity of the topics and lack of labels for those topics from the NLP analysis however is a challenge in implementation as we would need subject matter experts to intervene. Although this was not included in the final design, allowing users to change the number of topics in the results and adjusting the topic granularity could be a useful feature. We could also supply pre-defined labels for certain topics based on frequency throughout different users' datasets with the help of SMEs, and allow users to set their own labels for topics based on their colloquial use and understanding of certain vocabulary.

However, SME intervention adds more steps to processing the data, and it would be ideal to have an analysis flow that does not need human participation. To address this, we could potentially add back the word cloud to the graphic. Although this would complicate the overall visual, we could separate this the word clouds into a separate viewing mode. In the separate 'zoomed-in' view, when users select a specific topic bubble, we could change the bubble into a word cloud to represent the distributions within that selected topic. This would solve the need for SME provided labels without overcrowding the initial view of all the topics. Users would be able to clearly see which words are most prominent in the topic grouping, and they could infer their own labels from that information.

Fourth, most network maps are visually too busy and fail to offer useful insights. In our design, we made sure to avoid overwhelming our users with unnecessary information. The visual design we came up with allows for intuitive interactions and offers an actionable output that successfully answers how best to connect with a target contact. We foresee future design explorations to scale up with more advanced functionalities to find connections between multiple targets. And we also recognize calculating optimal connection paths as an area of algorithmic interest.

5 CONCLUSIONS

This paper provides a rich, insightful example of design problems and processes in the B2B context. More specifically, it reveals the challenges and opportunities around client interactions data capture and visualization, and offers details into design exploration of interaction visualizations, including requirements, related work, solutions, challenges, as well as reflections. Through the paper as a case study, it is hoped that the broad HCI community may benefit from both the design artifacts and the project experience.

Indeed, our project was not just a set of tasks and artefacts but a journey from identifying user needs, prototyping for feedback and realizing the limited prospect to pivoting to a related direction in hope of better reception and yet still bumping into major obstacles. While the project itself has certainly generated valuable intellectual capital, its direct impact is admittedly limited. In retrospect, we realized our project was a perfect example of innovation conundrum in the enterprise. That is, it's relatively straightforward to identify unfilled user needs of existing tools but really hard to introduce innovations into them, especially when they're SaaS solutions that have limited options for customization. In our case, the initial attempt to capture extra interaction data is blocked simply

because of the existence of the incumbent tool and the follow-up attempt to generate insights from the incumbent tool is hindered by its data limitation, which was targeted by the first attempt. It appears innovation is only possible when the entire incumbent system is replaced and such situation is quite rare. One lesson from our project for the HCI community is that meeting users' needs is often not enough in the B2B context because enterprise users lack the autonomy to choose many of the tools they use. Designers must consider multiple development trajectories, which should include integrating with or working around incumbent systems.

REFERENCES

- [1] Jonathan Grudin. 1988. Why CSCW Applications Fail: Problems in the Design and Evaluation of Organizational Interfaces. In *Proceedings of the 1988 ACM Conference on Computer-Supported Cooperative Work* (Portland, Oregon, USA) (CSCW '88). Association for Computing Machinery, New York, NY, USA, 85–93. <https://doi.org/10.1145/62266.62273>
- [2] Salesforce Labs. 2022. Time Warp - Salesforce Labs. <https://appexchange.salesforce.com/listingDetail?listingId=a0N3u00000ONm7EAD>
- [3] Radim Rehurek. 2022. Gensim: Topic modeling for humans. <https://radimrehurek.com/gensim/models/ldamodel.html>
- [4] E.R. Tufte. 2001. *The Visual Display of Quantitative Information*. Graphics Press. 8 pages. <https://books.google.com/books?id=GTd5oQEACAAJ>
- [5] Andrea Wagner. 2019. Real time: See Cradle to grave directly from agent timeline chart ... <https://brightmetrics.zendesk.com/hc/en-us/articles/360037343952-Real-Time-See-Cradle-to-Grave-Directly-from-Agent-Timeline-Chart>